DRAINCHECK / IMPLEMENTED WORKFLOW

How DrainCheck works

Real Sofia data, three separate analytical views, one layered map.

1 m GROUND INSPECTION
5 m ANALYSIS · VERSION 1.9

SOURCE DATA AND USER CHOICES	PROCESSING AND ASSUMPTIONS	MAPS AND COMPARISONS
01 TERRAIN **Municipal 1 m DTM** Real ground elevations Baltic 1982 height datum	REAL TERRAIN **Align to the 5 m grid** Elevation, depression fill depth, slope and contributing area.	RELATIVE INDICATOR **Calculate ponding signal P** 0–100 rank of positive fill depth. Zero where there is no depression.
02 EXPOSURE AND EVIDENCE **Past floods + buildings** 136 observed flood polygons Cadastral building footprints	CATEGORICAL RULES **Review terrain alongside actual flood evidence** Use P, observed flood footprints and potential building intersection. Historical categories take precedence over lower terrain categories.	EVIDENCE VIEW **Review priorities** Category map plus independent observed-flood overlays.
03 EDITABLE WHAT IF **Area, pipes and storm** Draw an area. Select existing pipes. Set rain, duration, blockage and assumed drainage removal. EVENT RAINFALL SLIDER	ASSUMED WATER BALANCE **Compare available drainage** Effective rain = rainfall × runoff coefficient **Baseline: full assumed removal** **Clogged: reduced removal** Capacity loss applies only inside the selected existing-pipe buffer. Excess runoff is measured in mm.	REUSES REAL 5 m TERRAIN **Estimate local ponding depth** Within each terrain depression, store only its locally generated excess runoff using actual bed elevations and fill depth. Cap storage at the spill level. Report overflow separately. Leave runoff on non-depression land unallocated.
		STORM COMPARISON **Baseline / clogged / change** **Estimated depth** m **Unremoved runoff** mm **Water volumes** m^3 Stored · overflow · unallocated

P

SHARED LAND EXCLUSIONS
Mapped water and uncertain terrain are excluded. Mapped greens fully absorb rain by the chosen assumption; source observations remain separate.

RIVER PROXIMITY CONTEXT
18 mapped river footprints + 19 assessment reaches → distance threshold → river-adjacent / away context. Proximity does not determine flood cause.

SUPPLEMENTAL MAP LAYERS
Archived daily rain, official assessments and infrastructure inventory remain separate context. Archived rain does not drive the storm slider.

Comparative screening, not a calibrated flood forecast. Scenario omits dynamic flow routing, pipe hydraulics and river-level simulation.

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